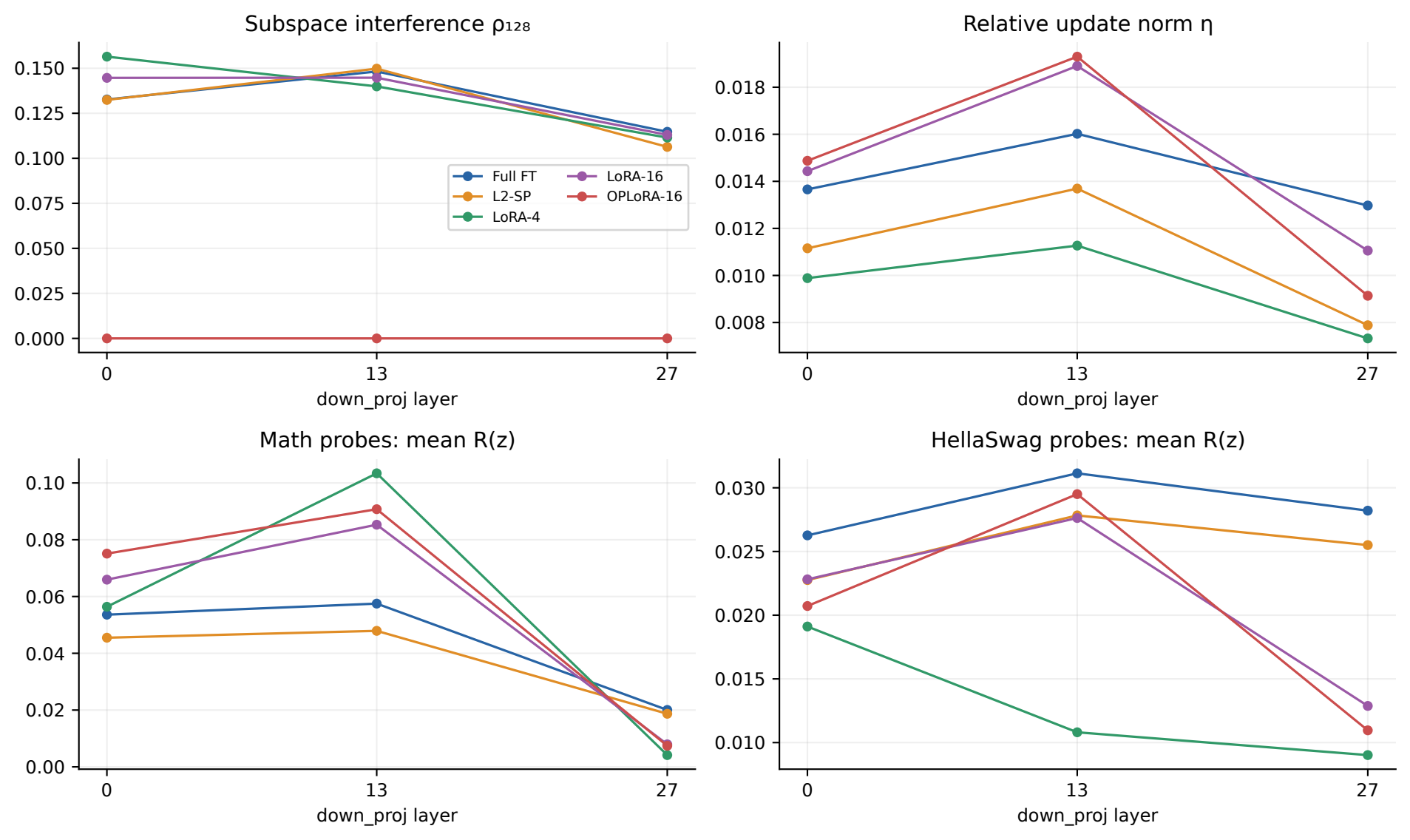
Subspace interference ρ128
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
13
27
down_proj layer
Full FT
L2-SP
LoRA-4
LoRA-16
OPLoRA-16
Relative update norm η
0.018
0.016
0.014
0.012
0.010
0.008
0
13
27
down_proj layer
Math probes: mean R(z)
0.10
0.08
0.06
0.04
0.02
0.00
0
13
27
down_proj layer
HellaSwag probes: mean R(z)
0.030
0.025
0.020
0.015
0.010
0
13
27
down_proj layer